AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Physics Paper II

Time: 1 hour 30 minutes Marks: 35

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.

3. There are ELEVEN questions in this exam. Answer ALL questions. Questions 10 & 11 each offer TWO choices. For each of these questions, attempt ONE choice only.

4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.

5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.

6. DO NOT mention your name or school name anywhere inside the paper.

7. You may use a scientific calculator if you wish

Q.1. (Total 3 Marks)

Two identical objects are placed in vacuum with their centres 2.10 m apart. The gravitational force of attraction between them is 0.39 N.

(**Note:** The value of universal gravitational constant is $6.673\times10^{-11}\ Nm^2/kg^2$.)

i. Calculate the mass of each object. (2 Marks)

ii. Without further calculation, how will the force change if the distance between the objects becomes doubled? (1 Mark)

Q.2. (Total 3 Marks)

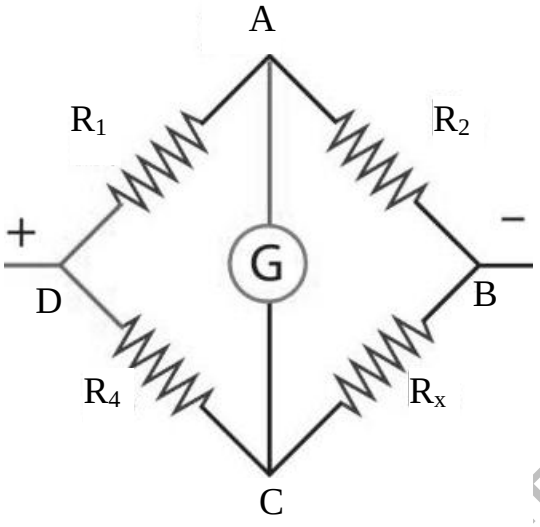
A parallel plate capacitor is connected to a battery. If these plates are moved further apart, then work will be done against the electrostatic attraction between the plates.

Describe the effect of moving the plates of the capacitor on its stored energy.

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Q.3. (Total 3 Marks)

The given diagram shows an electrical circuit.



i. Identify the name of the circuit. (1 Mark)

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ii. To obtain null condition in the galvanometer connected between points A and C, which two points of the circuit must be at the same potential? (1 Mark)

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iii. Write the mathematical relation among the four resistances of the circuit under the null condition. (1 Mark)

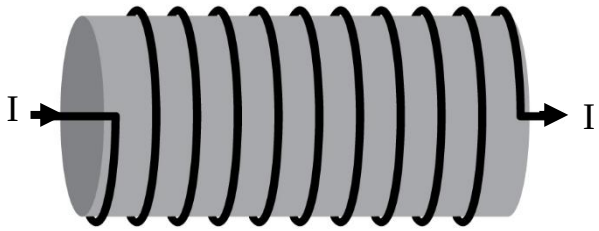
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Q.4. (Total 4 Marks)

The given diagram shows a component of an electrical device.



i. Draw a diagram to show the direction of current [using dot (•) and cross (×) notation] and the direction of magnetic field around each turn of the component. (2 Marks)

Space for diagram

ii. By naming the appropriate rule, identify the direction of the magnetic field inside the solenoid. (2 Marks)

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Q.5. (Total 2 Marks)

i. Mention the name of the instrument used to detect earthquakes. (1 Mark)

_____

_____

ii. State any ONE other use mentioned in part (i) of this instrument. (1 Mark)

_____

_____

Q.6. (Total 2 Marks)

According to Louis deBrogli, particles in motion exhibit wave-like behaviour.

i. What happens to the deBrogli wavelength of a particle when its speed increases? (1 Mark)

_____

_____

ii. Why is the wave nature of particles not noticeable for macroscopic objects? (1 Mark)

_____

_____

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Q.7. (Total 3 Marks)

For the nucleus of $^{46}_{22}Ti$.

i. Write the mathematical expression for the mass defect by showing the number of protons and neutrons. (2 Marks)

ii. Convert the mass defect expression into the binding energy equation. (1 Mark)

Q.8. (Total 2 Marks)

Describe a reason for each of the following statement.

i. Strong nuclear force effective only at very short distances. (1 Mark)

ii. The gravitational force dominates at astronomical scales. (1 Mark)

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Q.9. (Total 3 Marks)

i. Describe Hubble's law. (1 Mark)

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ii. How does the given law provides evidence in favour of the Big Bang theory? (2 Marks)

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Q.10. (Total 5 Marks)

EITHER

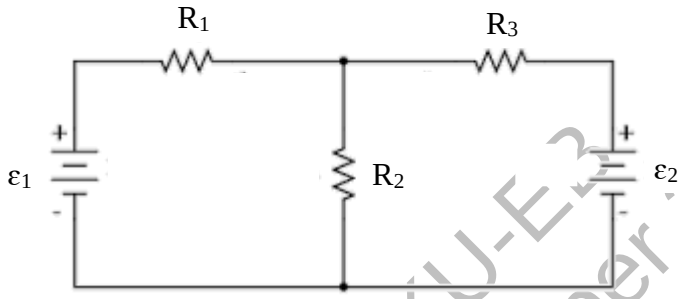
a.

i. Describe the significance of Gauss's law. (1 Mark)

ii. Derive the mathematical expression for the electric field intensity in terms of surface charge density for an infinite plane sheet of uniform charge, using Gauss's law. (4 Mark)

OR

b. The given circuit consists of three resistors of resistances, 6 Ω, 3 Ω and 9 Ω respectively and two sources of 6 V and 4 V respectively.



Calculate the current flowing through each resistor using Kirchhoff's law. (5 Marks)

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Q.11. (Total 5 Marks)

EITHER

a.

i. Why is the average value of an alternating current over a complete cycle zero? (1 Mark)

ii. Elaborate the importance of using the root mean square (r.m.s) value of A.C. and relate the r.m.s value of A.C to an equivalent D.C value. Also write its mathematical expression. (4 Marks)

OR

b.

i. Describe Compton's effect and write the evidence provided by Compton effect for the particle nature of electromagnetic radiation. (2 Marks)

ii. Relate the change in wavelength of scattered X-rays to the interaction between radiation and matter in THREE points. (3 Marks)

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END OF PAPER

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